

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

PRACTICAL - 14

AIM : Extracting date components using lubridate:: functions (R).

Input :

```
# R Script: Extracting Date Components using lubridate
# Dataset: Student Attendance
# Load necessary libraries
install.packages("lubridate")
# install.packages("dplyr")
library(lubridate)
library(dplyr)
# 1. SETUP: Create Sample Student Attendance Data
# We start with attendance dates stored as character text (YYYY-MM-DD)
attendance_df <- data.frame(
  Student_ID = 1:4,
  Attendance_Date = c("2023-06-15", "2023-10-31", "2024-02-29", "2024-12-01")
)
print("--- 1. Original Student Attendance Data ---")
print(attendance_df)

# 2. PARSE AND EXTRACT DATE COMPONENTS
processed_data <- attendance_df %>%
  mutate(
    # A. Parsing: Convert text to Date (Year-Month-Day)
    Actual_Date = ymd(Attendance_Date),
    # B. Extraction Functions
    Year_Num = year(Actual_Date),
    Month_Num = month(Actual_Date),
    Month_Name = month(Actual_Date, label = TRUE),
    Day_Num = day(Actual_Date),
    Weekday_Num = wday(Actual_Date),
    Weekday_Name = wday(Actual_Date, label = TRUE, abbr = FALSE),
    Quarter = quarter(Actual_Date),
    Day_of_Year = yday(Actual_Date)
  )
print("--- Data with Extracted Date Components (Student Attendance) ---")
print(processed_data)
```

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3. System Date: Handling "Now"

Extracting components from the current system timestamp

```
current_time <- now()
```

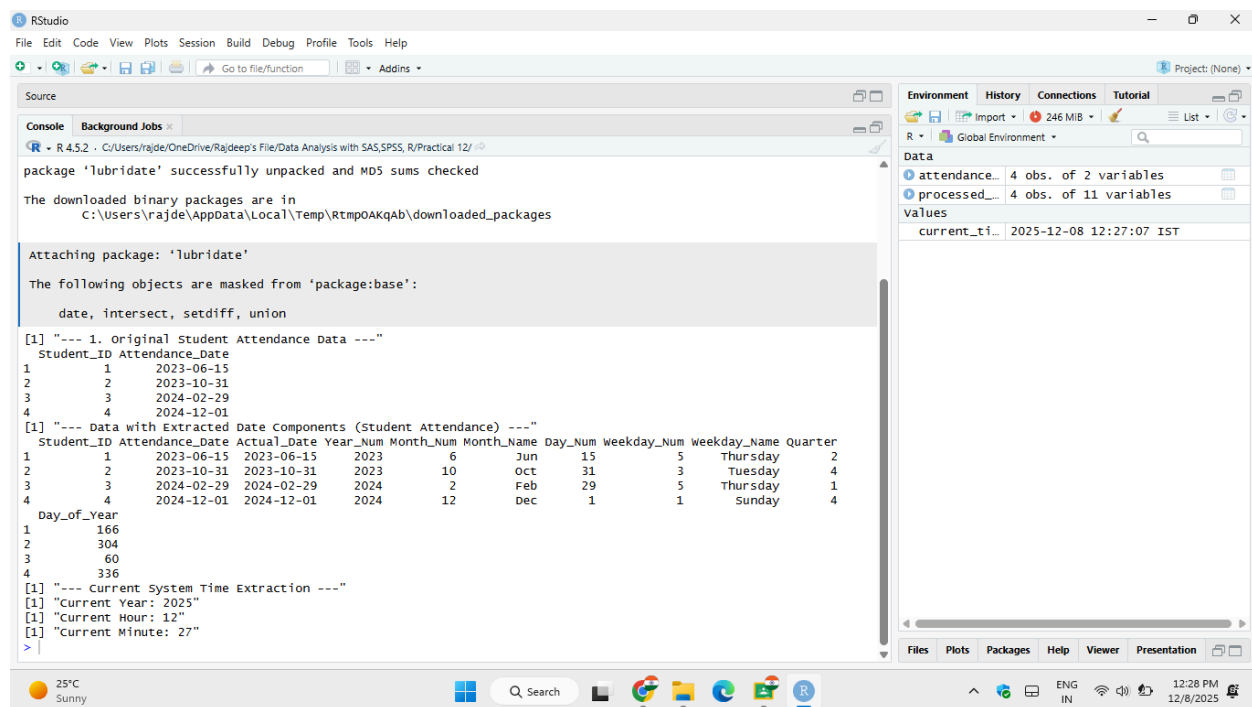
```
print("--- Current System Time Extraction ---")
```

```
print(paste("Current Year:", year(current_time)))
```

```
print(paste("Current Hour:", hour(current_time)))
```

```
print(paste("Current Minute:", minute(current_time)))
```

Output :



The screenshot shows the RStudio interface with the following console output:

```
R - R 4.5.2 - C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 12/
package 'lubridate' successfully unpacked and MD5 sums checked
The downloaded binary packages are in
  C:\Users\rajde\AppData\Local\Temp\RtmpOAKqAb\downloaded_packages

Attaching package: 'lubridate'

The following objects are masked from 'package:base':
  date, intersect, setdiff, union

[1] "--- 1. Original Student Attendance Data ---"
Student_ID Attendance_Date
1          1      2023-06-15
2          2      2023-10-31
3          3      2024-02-29
4          4      2024-12-01
[1] "--- Data with Extracted Date Components (Student Attendance) ---"
Student_ID Attendance_Date Actual_Date Year_Num Month_Num Month_Name Day_Num weekday_Num weekday_Name Quarter
1          1      2023-06-15      2023-06-15      2023         6      Jun         15         5      Thursday      2
2          2      2023-10-31      2023-10-31      2023        10      Oct         31         3      Tuesday      4
3          3      2024-02-29      2024-02-29      2024         2      Feb         29         5      Thursday      1
4          4      2024-12-01      2024-12-01      2024        12      Dec          1         1      Sunday       4
Day_of_Year
1          166
2          304
3           60
4          336
[1] "--- Current System Time Extraction ---"
[1] "Current Year: 2025"
[1] "Current Hour: 12"
[1] "Current Minute: 27"
>
```

The Environment pane on the right shows the following data:

Object	Variables
attendance...	4 obs. of 2 variables
processed...	4 obs. of 11 variables
current_ti...	2025-12-08 12:27:07 IST

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